Result 1 (restore a restricted element). [Subtype restriction, original element round trip]
Let A be a type. Let S and T be subsets of A. Let U be a subset of S. Let *support* be evidence identifying T with the elements of S selected by U. Let x be an element of T. Then

$$j(r(x)) = x.$$

Result 2 (restrict a restored element). [Subtype restriction, selected element round trip]
Let A be a type. Let S and T be subsets of A. Let U be a subset of S. Let *support* be evidence identifying T with the elements of S selected by U. Let y be an element of U. Then

$$r(j(y)) = y.$$

Result 3 (supported sets correspond). [Subtype restriction, value preserving equivalence]
Let A be a type. Let S and T be subsets of A. Let U be a subset of S. Let *support* be evidence identifying T with the elements of S selected by U. Then

$$(\text{there exists } e : T \simeq U,\ (\text{for every } x : T,\ e(x) = x))\,.$$

LLM-assisted statements and proofs (gpt-6, 2026-11-07). See PROVENANCE.md. These generic results require the explicitly stated support record. Constructing that record for interval labels remains a separate check.